Validation and AI Use — Project 1 Working Source

Decision, version, and as-of boundary

- **Decision/user:** Watch/defer Apple for a buy-side investment committee with no current position.
- **Source cut-off:** September 17, 2026 for market-price comparison; FY2025 audited statements for opening balances; Q3 FY2026 Form 10-Q for the bridge.
- **Environment:** Python standard library only; Python 3.9+ documented in README.
- **Repository:** https://github.com/catiemoe-cloud/Project-1-

Validation register

Claim/output	Failure mode	Test / expected result	Actual result	Disposition	Evidence
Three-statement articulation	DCF may calculate while balance sheet/cash do not tie	Run `python3 src/apple_fcff_model.py`; every gap/tie must be 0.00	FY2026E–FY2030E each printed 0.00	Pass	`src/apple_fcff_model.py`; `docs/cold-run.md`
FCFF calculation	Financing could leak into unlevered cash flow	Inspect NOPAT + D&A − capex − change NWC	Financing/capital return excluded from FCFF	Pass	`src/apple_fcff_model.py`
Enterprise-to-equity bridge	Cash/debt could be omitted or double counted	EV + $146,517M − $84,344M, divided by 14,594.18M shares	Base EV $1,858,503.5M; equity $1,920,676.5M; $131.61/share	Pass	`src/analysis.py`; Q3 FY2026 10-Q
Product guardrails	Incoherent inputs could show a false value	Render with WACC 5%; expected error	Returned “Input check failed”	Pass	`src/app.py`; `docs/cold-run.md`
Known answer	Code change could silently alter base case	Base model must return $131.61/share	Returned $131.61/share	Pass	`output/visible_output.json`
Published training known answer	DCF mechanics could be wrong despite an Apple output	Week 3 case must return $27.50/share; 11% WACC probe about $23.41	$27.50 and $23.41	Pass	`src/apple_fcff_model.py`; Week 3 worked example
DCF monotonicity and boundary	Sensitivity could move in the wrong direction or value an invalid terminal rate	WACC up must reduce value; terminal growth up must increase value; every grid cell has g < WACC	Passed; 8%–10% WACC and 2.0%–3.0% g grid	Pass	`src/apple_fcff_model.py`; `output/visible_output.json`
WACC component build	Discount rate could be an unsupported scalar	Recompute CAPM/market-value build from dated risk-free rate, beta, ERP, debt cost, tax rate, and capital weights	9.45% calculated; 9.0% model input remains within the documented 0.5 percentage-point whole-percent rounding band	Pass, with limitation	`docs/wacc-build.md`; `src/apple_fcff_model.py`
Equity multiple	Peer P/E could be inconsistent	Manual MSFT check: $497.75 ÷ $17.95	27.729805x	Pass	`src/peer_valuation.py`
EV multiple	EV/revenue could mix definitions	Alphabet EV = price × shares − cash/securities + debt	$4,340,432.26M / $402,836M = 10.774688x	Pass with comparability limitation	`research/peer-policy.md`
README-only cold run	Hidden dependencies/steps	Fresh directory; execute documented commands	First attempt passed in under one second	Pass	`docs/cold-run.md`

Locked Changed-Input Record

The prediction below was supplied by the student. The baseline commit was created after that prediction timestamp and immediately before the run; this timing is disclosed rather than backdated.

- **Human AI-off prediction, written before changing the input:** Before changin

g the model, I predict that increasing annual revenue growth from 6.0% to 9.0%, while keeping all other factors the same, will increase Apple’s projected revenue, EBIT, FCFF, and DCF value per share. However, I do not think this will change my recommendation because the original valuation is still much lower than the current market price of $337.00.
- **Prediction timestamp:** October 9, 2026, 1:43 PM ET. I wrote the prediction on that date but did not create the baseline Git commit until October 16, 2026.
- **Baseline commit:** `0ba92e5`, created October 16, 2026, immediately before the changed-input run. It corroborates the run state but does not represent an October 9 pre-run commit.
- **Changed input:** Revenue growth, old 6.0%, new 9.0%; units: annual percentage of prior-year revenue.
- **Expected direction:** Higher revenue growth should increase forecast FCFF and value per share, holding all other model inputs fixed.
- **Run command:** `python3 src/apple_fcff_model.py` after making the one input change, or use the product’s 9.0% revenue-growth control and press “Run valuation.”
- **Actual result:** Base value was $131.61/share. With only revenue growth changed to 9.0%, the value increased to $148.85/share. FY2026E revenue / EBIT / FCFF increased from $441,130.66M / $131,195.01M / $107,528.53M to $453,615.49M / $135,190.16M / $110,805.83M. FY2030E revenue / EBIT / FCFF increased from $556,917.29M / $165,448.90M / $135,781.38M to $640,315.28M / $191,414.74M / $156,318.22M. Base and changed cases passed every balance-sheet and cash-flow articulation check.
- **Decision effect:** Watch/defer did not change because the changed value, $148.85/share, remained below the $337.00 market price.
- **Diagnosis:** Higher revenue growth increased forecast revenue. With gross margin, operating expenses, capex, WACC, and terminal growth held fixed, the higher revenue path increased EBIT and FCFF; discounting that larger FCFF stream increased enterprise value and value per share.

The changed-input record is complete. Freeze the PDF only after confirming that the Markdown and PDF match exactly.

Named AI-use register

Tool / surface	Interaction date	Material task	Independent check	Disposition
Codex / generative AI	Oct. 2026	Organized documentation, reviewed requirements, and assisted with code drafting and source reconciliation.	The model has executable checks, primary-filing links, and a cold run; student remains responsible for assumptions and final locked prediction.	Disclosed development aid.

Limitations and kill rule

Do not rely on the valuation if the source cut-off, bridge convention, base-case assumptions, or articulation checks cannot be reproduced. The forecast aggregates operating assets and liabilities instead of forecasting every Apple line item. The market-multiple work is a non-averaged cross-check, not proof that the DCF result is correct.